Step 7 − composition and machine learning (bacteria only, not rarefied)

Bacterial ASVs: 2286 Samples: 32 canker + 8 healthy
Healthy samples dropped (0 bacterial reads after chloroplast removal):
CP2_Healthy, Fuji1_Healthy, Fuji3_Healthy, Fuji4_Healthy, Fuji5_Healthy, HW1_Healthy, HW2_Healthy, Nic1_Healthy

ASV IDs are ours (ranked by abundance); see results/07_composition_ml/asv_lookup.tsv

Fig. 2A reproduction: class−level composition (bacteria only)

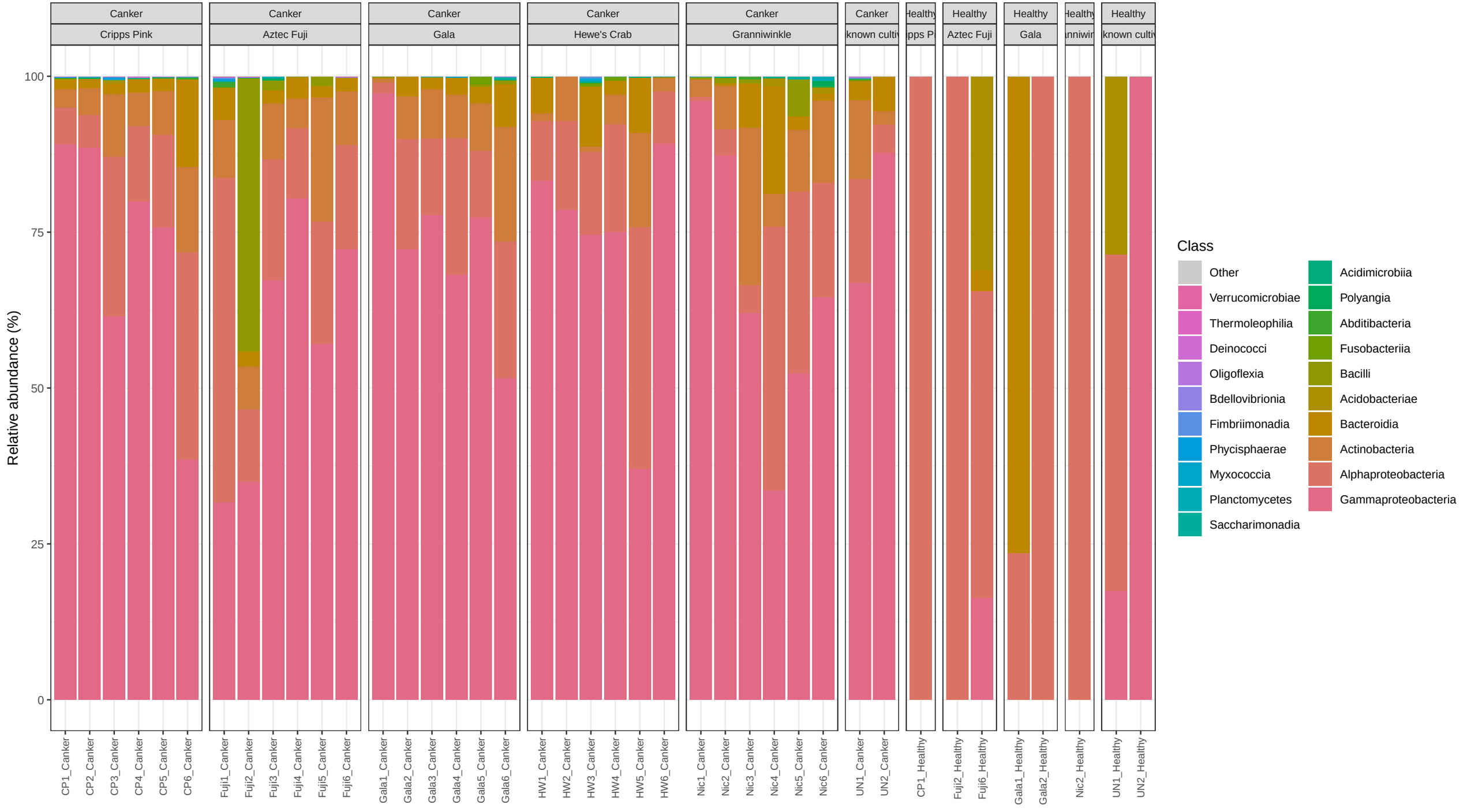
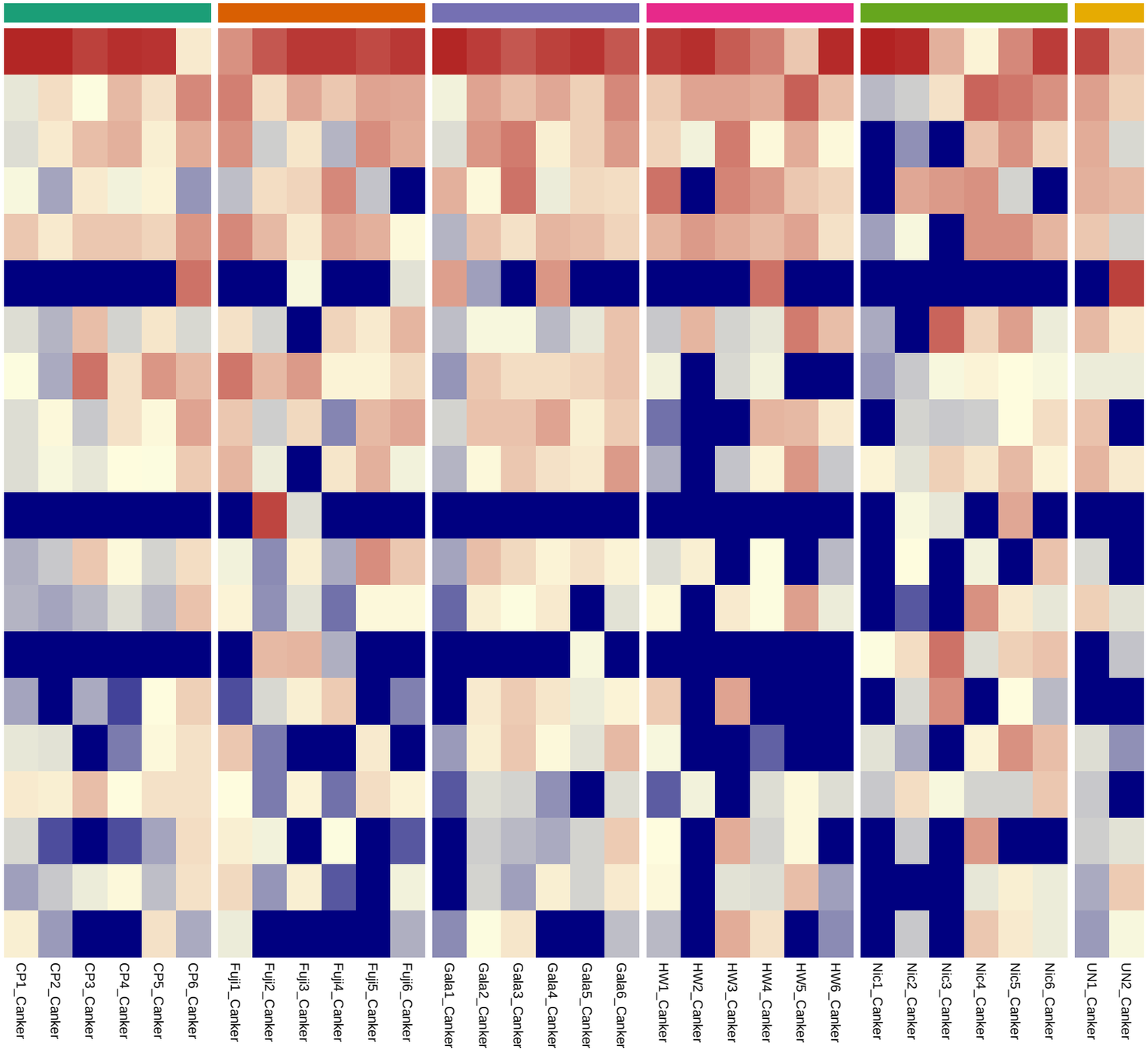
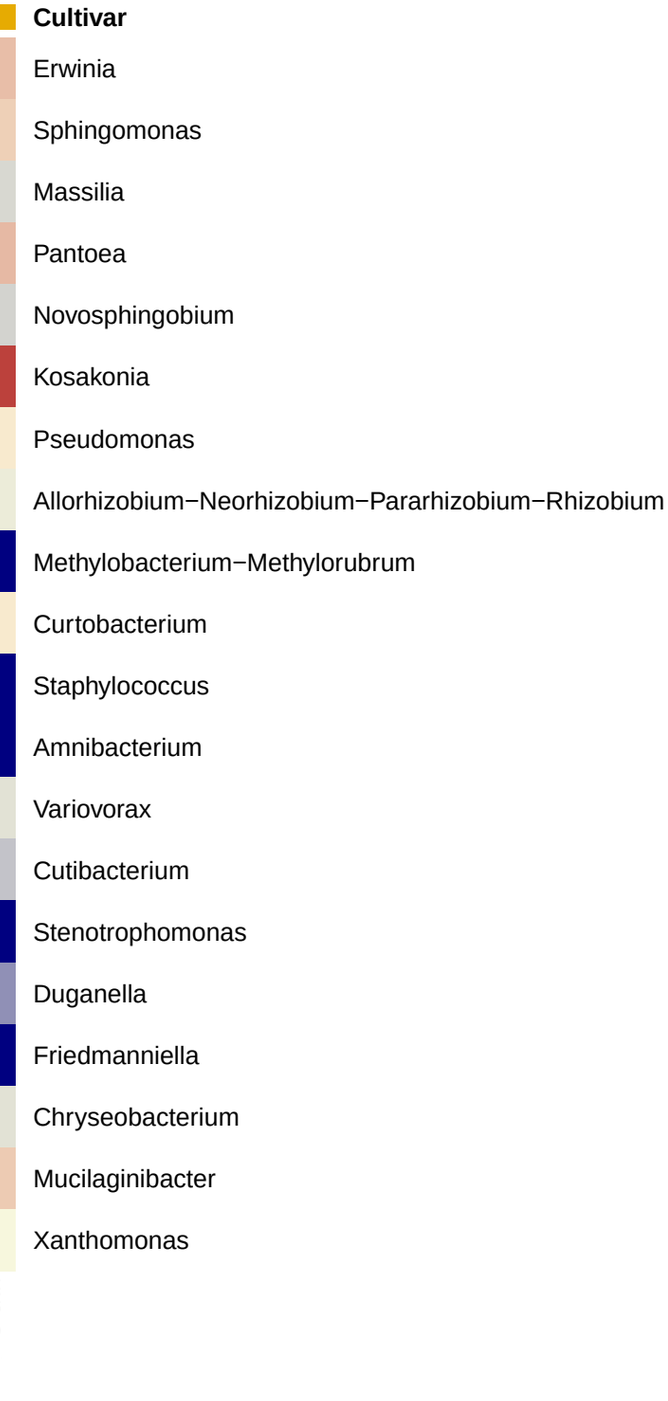
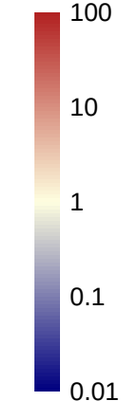
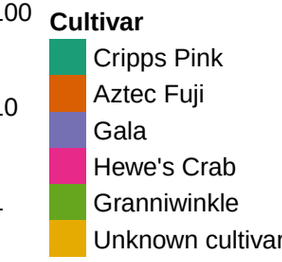


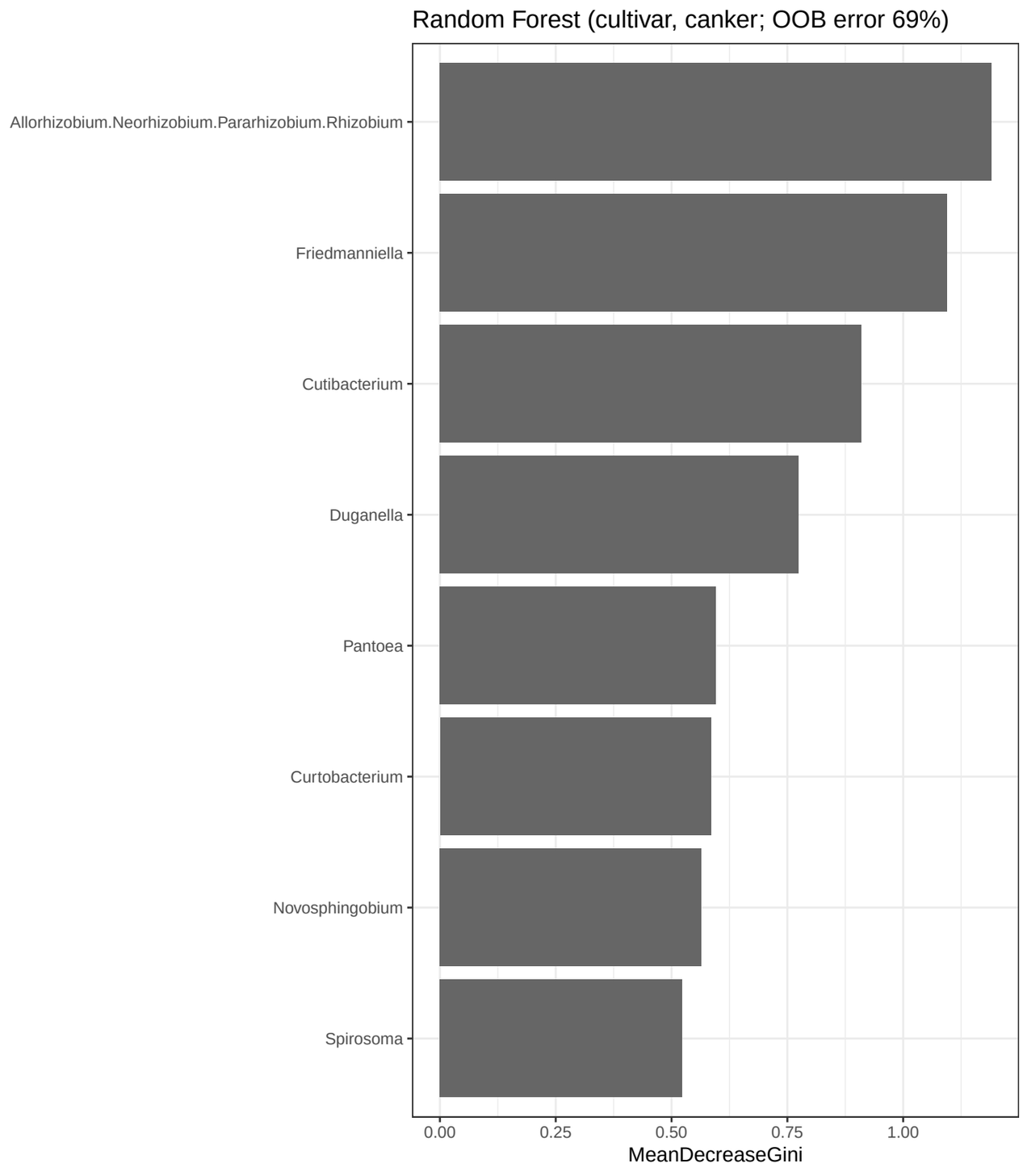
Fig. 4A reproduction: top 20 genera in canker samples (% relative abundance, log scale)

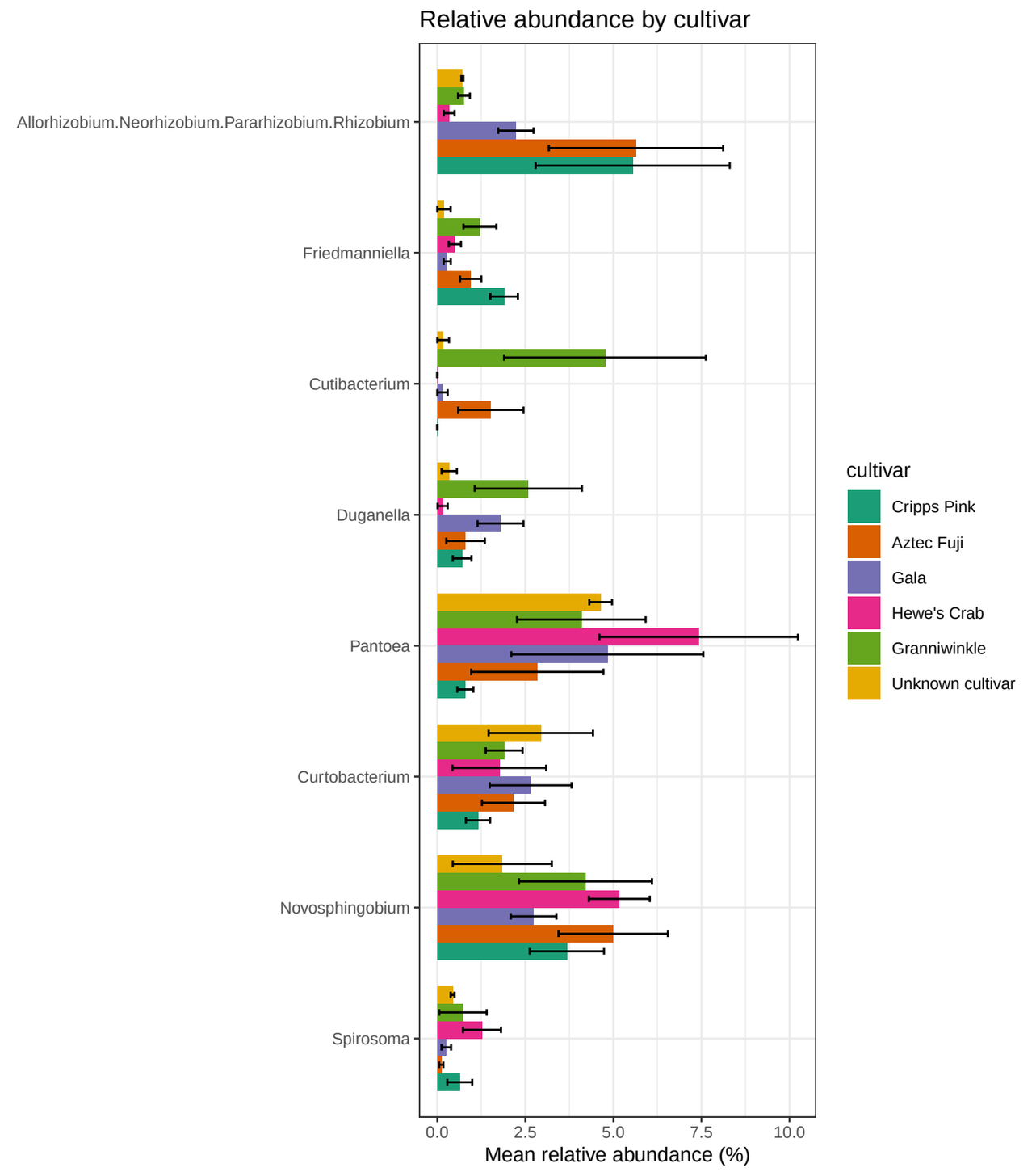









Random Forest (cultivar, canker; OOB error 69%)
Allorhizobium.Neorhizobium.Pararhizobium.Rhizobium
Friedmanniella
Cutibacterium
Duganella
Pantoea
Curtobacterium
Novosphingobium
Spirosoma
0.00
0.25
0.50
0.75
1.00
MeanDecreaseGini


Relative abundance by cultivar
Allorhizobium.Neorhizobium.Pararhizobium.Rhizobium
Friedmanniella
Cutibacterium
Duganella
Pantoea
Curtobacterium
Novosphingobium
Spirosoma
0.0
2.5
5.0
7.5
10.0
Mean relative abundance (%)
cultivar
Cripps Pink
Aztec Fuji
Gala
Hewe's Crab
Granniwinkle
Unknown cultivar

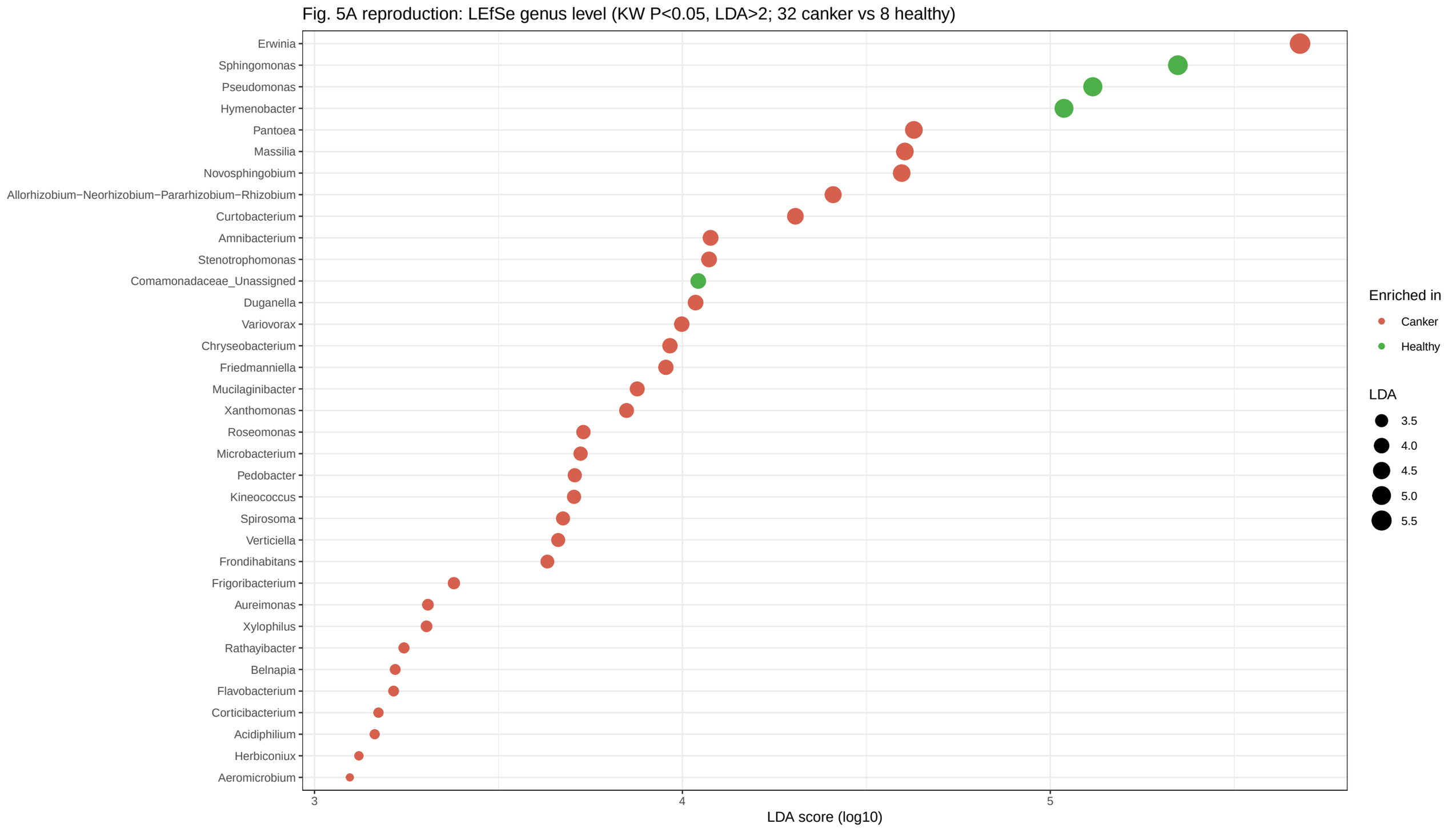


Fig. 5A reproduction: LEfSe genus level (KW P<0.05, LDA>2; 32 canker vs 8 healthy)
Erwinia
Sphingomonas
Pseudomonas
Hymenobacter
Pantoea
Massilia
Novosphingobium
Allorhizobium−Neorhizobium−Pararhizobium−Rhizobium
Curtobacterium
Amnibacterium
Stenotrophomonas
Comamonadaceae_Unassigned
Duganella
Variovorax
Chryseobacterium
Friedmanniella
Mucilaginibacter
Xanthomonas
Roseomonas
Microbacterium
Pedobacter
Kineococcus
Spirosoma
Verticiella
Frondihabitans
Frigoribacterium
Aureimonas
Xylophilus
Rathayibacter
Belnapia
Flavobacterium
Corticibacterium
Acidiphilium
Herbiconiux
Aeromicrobium
3
4
5
LDA score (log10)
Enriched in
Canker
Healthy
LDA
3.5
4.0
4.5
5.0
5.5

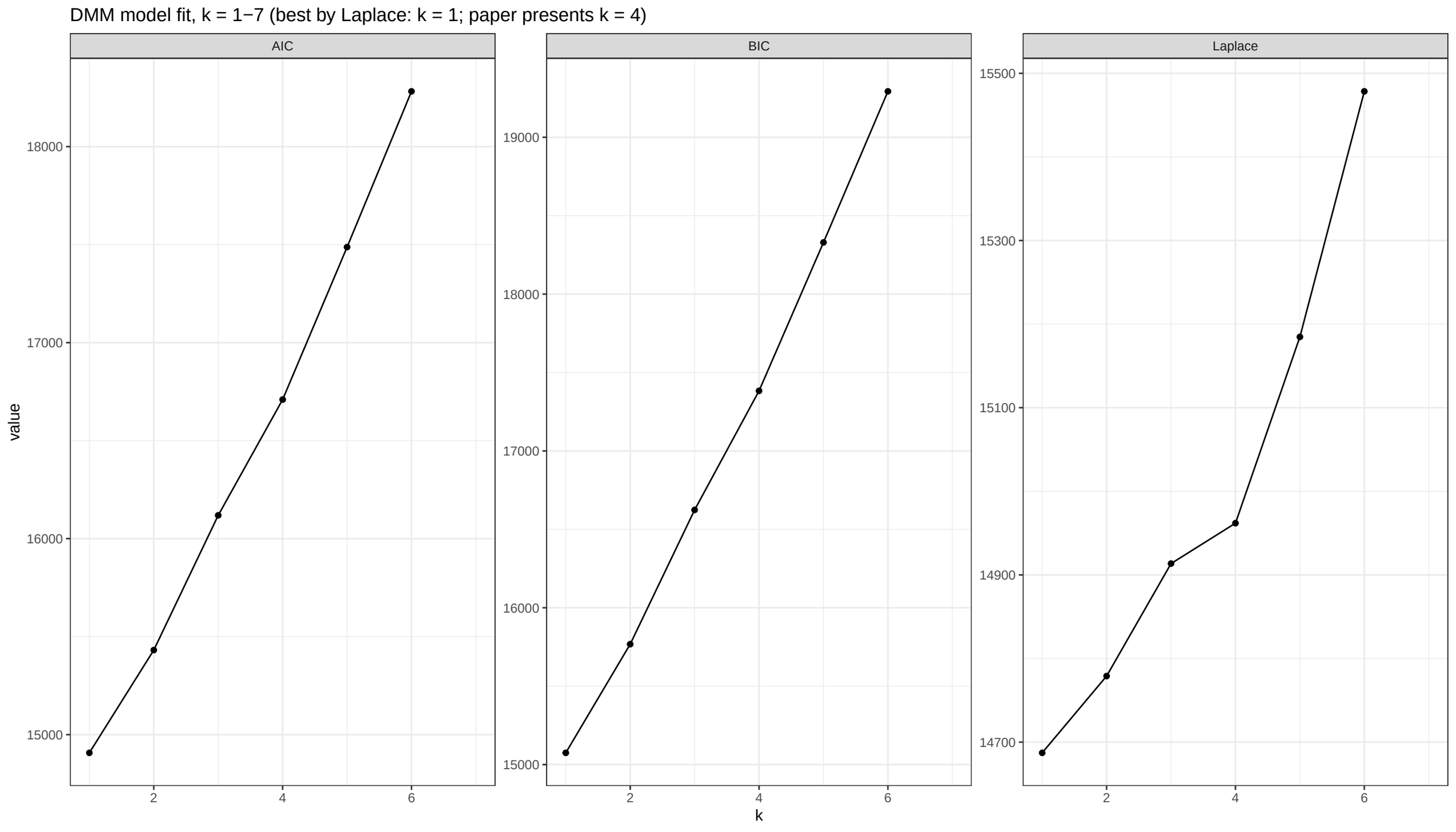

DMM model fit, k = 1−7 (best by Laplace: k = 1; paper presents k = 4)
AIC
BIC
Laplace
value
k
18000
17000
16000
15000
19000
15500
15300
15100
14900
14700
2
4
6

Fig. 5D reproduction: top 10 drivers per DMM community type (k = 4)

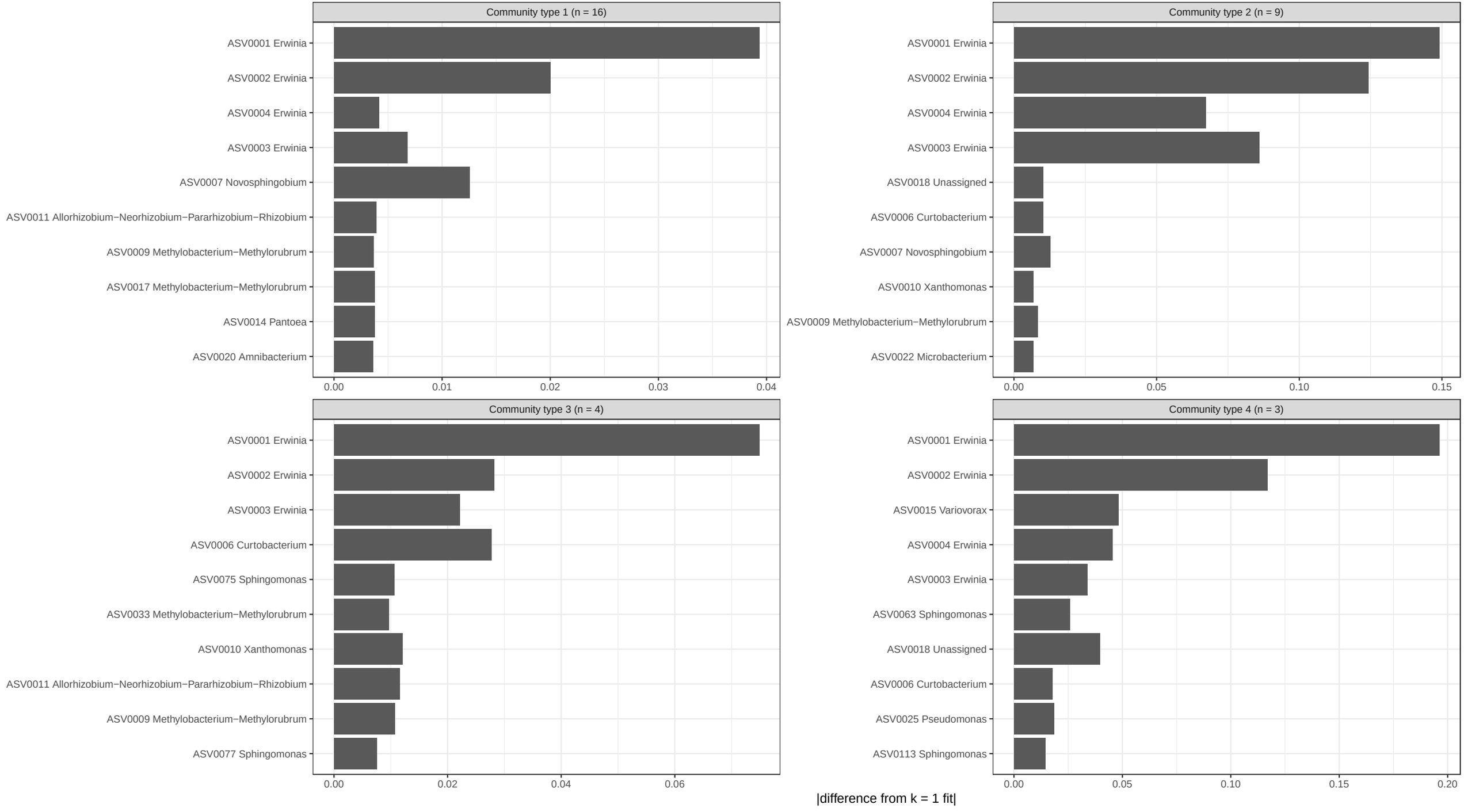


Fig. 5B reproduction: top 100 ASVs (sqrt counts) in canker samples ordered by DMM community type (k = 4)

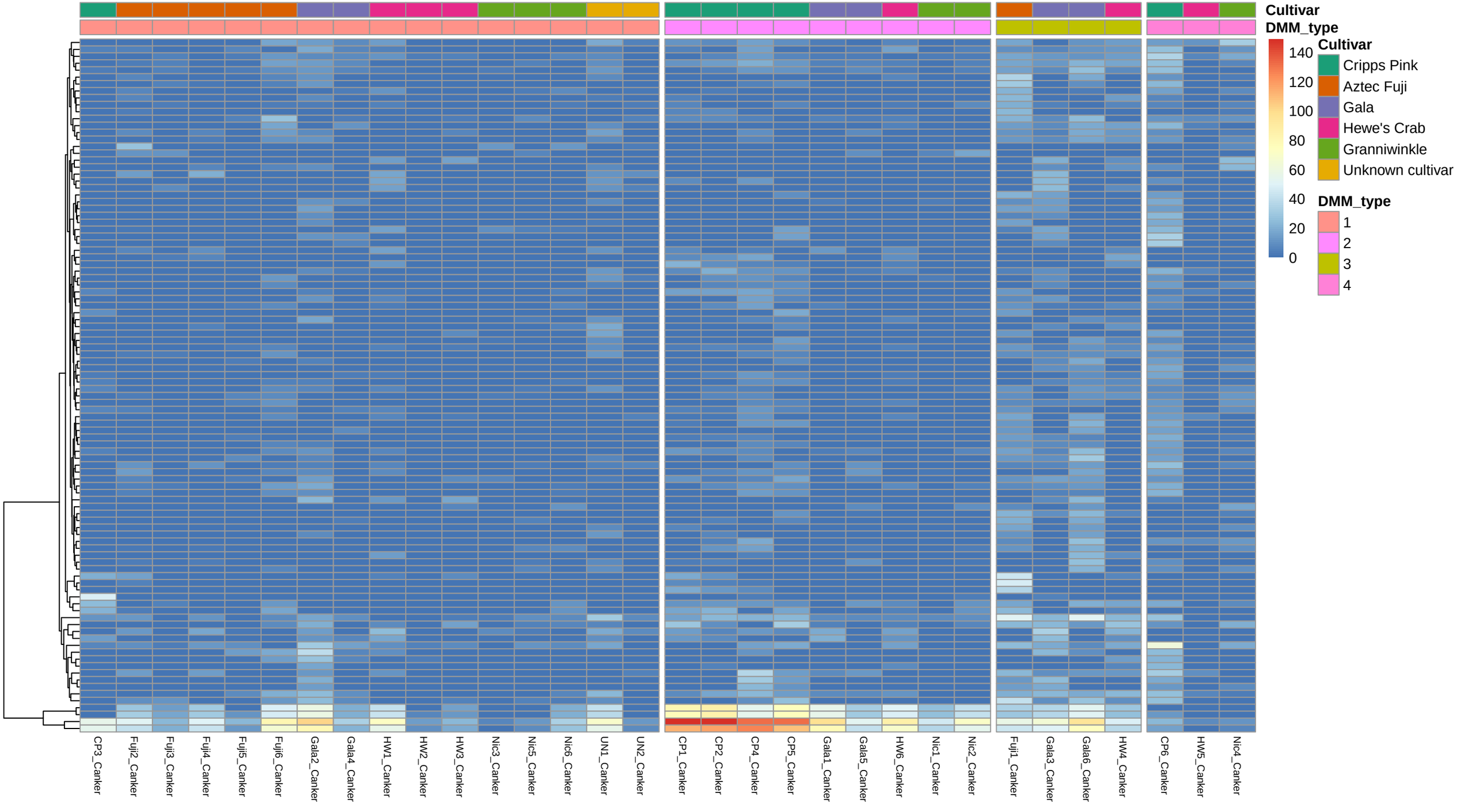
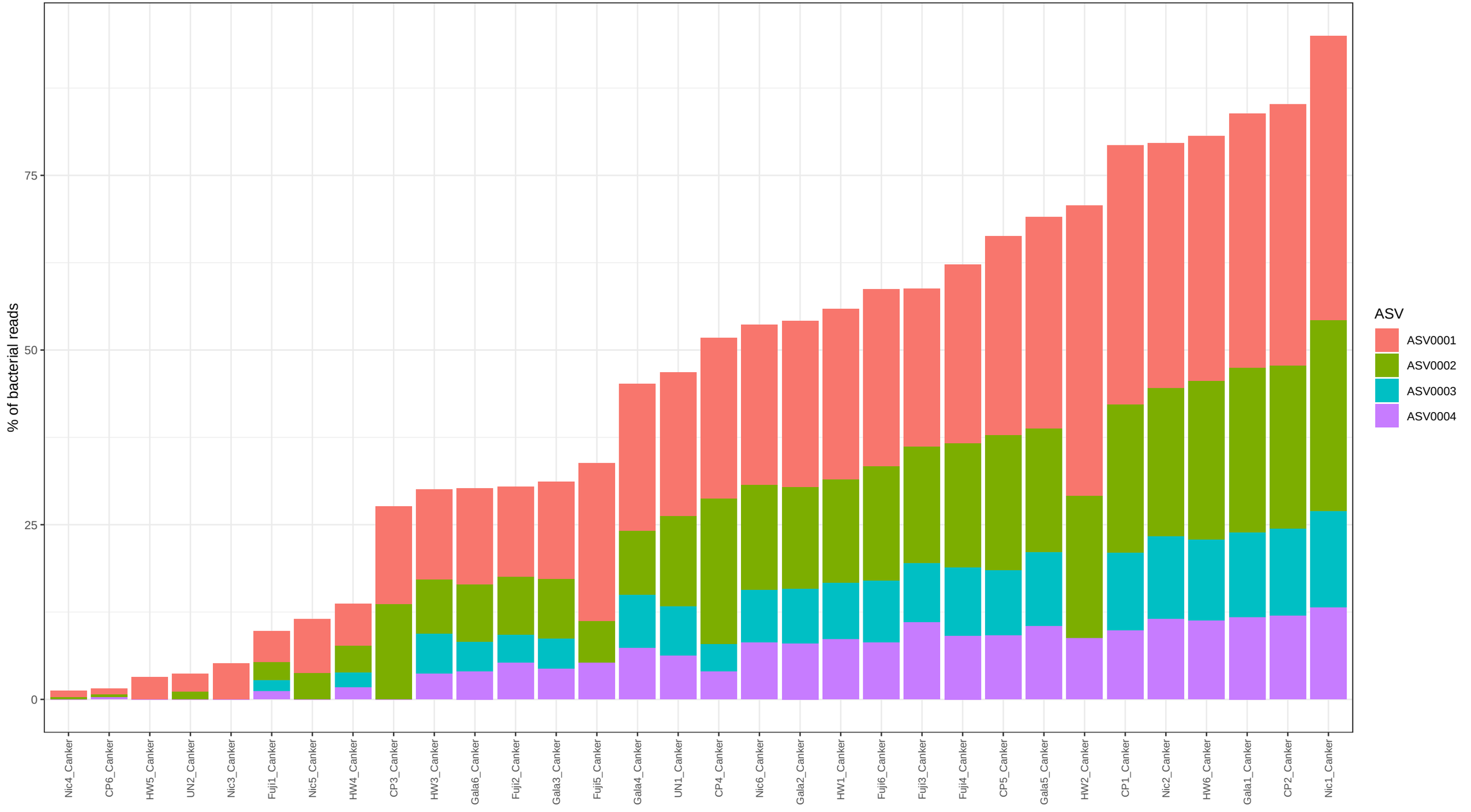


Fig. 5C reproduction: the four most abundant Erwinia ASVs per canker sample



Erwinia ASVs by cultivar (paper ranking: Cripps Pink > Gala > Aztec Fuji > Granniwinkle > Hewe's Crab > Unknown)

cultivar	samples	total_Erwinia_ASVs	mean_ASVs_per_sample	mean_Erwinia_pct	mean_Shannon
Cripps Pink	6	10	5.00	60.0	1.322
Gala	6	8	4.83	53.3	1.319
Aztec Fuji	6	6	4.17	43.4	1.269
Hewe's Crab	6	4	3.33	42.3	1.006
Granniwinkle	6	4	2.83	41.0	0.840
Unknown cultivar	2	4	3.00	25.2	0.942